Answer **one** question in this section in the spaces provided.

7 Fig. 7.1 shows a small mass attached to the end of a fixed spring.

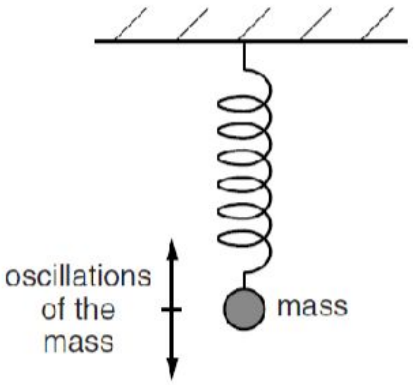


Fig. 7.1 (not to scale)

The mass is given a small vertical displacement and then released. It oscillates with simple harmonic motion about its equilibrium position.

(a) Define the term *simple harmonic motion*.

acceleration is proportional to displacement from fixed point [1]

acceleration directed to fixed point (equilibrium point) [1]

.

.....

..... [2]

(b) The graph in Fig. 7.2 shows how the displacement x varies with time t for the motion of the mass on the spring.

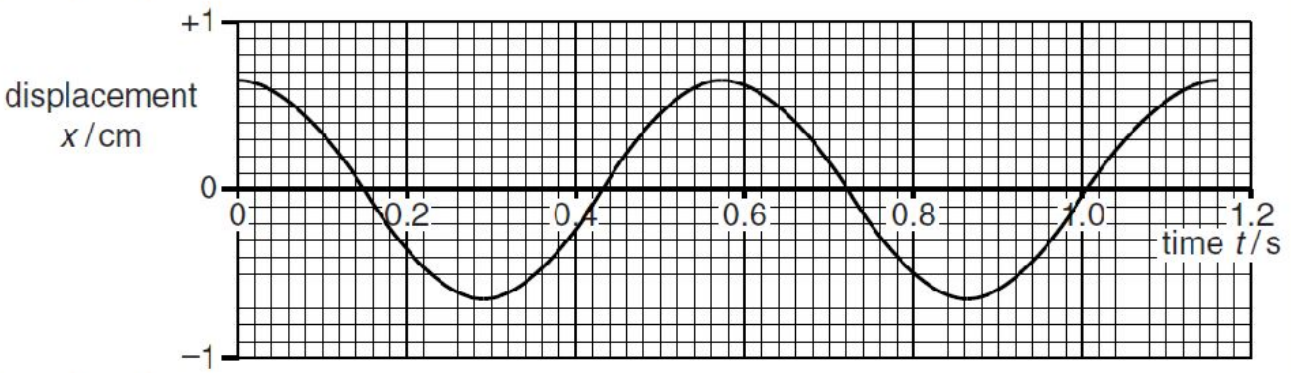


Fig. 7.2

(i) Using data from the graph in Fig. 7.2, determine the velocity of the mass at time $t = 0.66$ s.

(from graph in Fig. 7.2)

$T = 0.58$ (s)

$A = 0.65$ (cm) [1]

$\omega = 2\pi/T$ [1]

substitution $v = -A\omega \sin \omega t = -0.65 \times 10.83 \sin (10.83 \times 0.66)$ [1]

$= -5.4$ cm s^{-1}

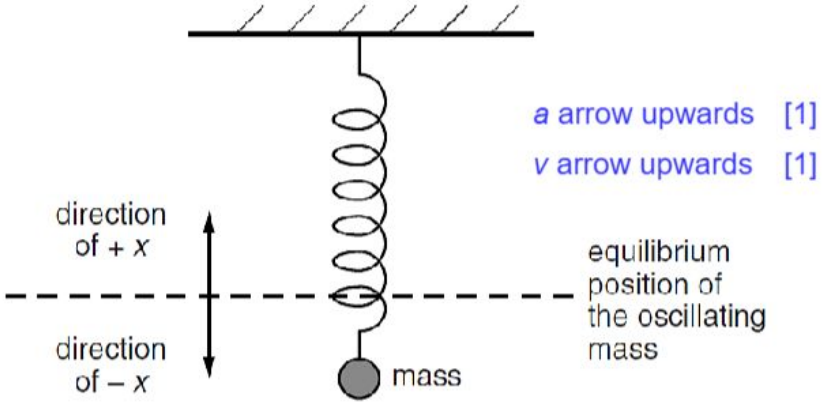
OR

the gradient of tangent at $t = 0.66$ s is negative, so

$$v = -\omega\sqrt{x_0^2 - x^2} = -\frac{2\pi}{0.58}\sqrt{0.65^2 - 0.40^2} = -5.6 \text{ cm s}^{-1}$$

velocity = ... cm s^{-1} [3]

(iii) Fig. 7.3 shows the spring and the position of the mass at instant $t = 0.94$ s.



a arrow upwards [1]

v arrow upwards [1]

Fig. 7.3

On Fig. 7.3, add arrows labelled *a* and *v* to show the directions of the acceleration and the velocity, respectively, at this instant in time. [2]

(c) Fig. 7.3 shows a uniform rod floating vertically in water of density ρ. The rod has a mass M and a uniform cross-sectional area A.

When the rod is in equilibrium, its base is at a depth h below the water surface.

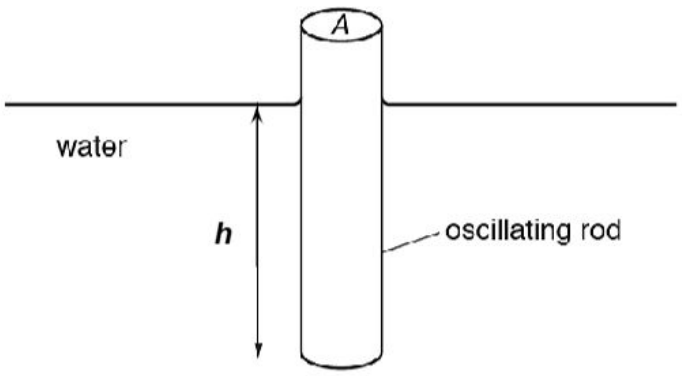


Fig. 7.3

The rod is displaced vertically and released. The acceleration a of the resulting oscillations is given by

$$a = -\left(\frac{\rho A g}{M}\right) x$$

where x is the displacement from the equilibrium depth h and g is the acceleration due to free fall.

(i) Show that the motion of the rod is simple harmonic.

The given equation is:

$$a = -\left(\frac{\rho A g}{M}\right) x$$

Since ρ, M, A and g are constants, the acceleration is directly proportional to displacement x and is in opposite direction to x (indicated by the negative sign), the motion of the rod is simple harmonic. [1]

---

the motion is simple harmonic, with:

$$\omega^2 = \frac{\rho A g}{M}$$

[1]

[2]

(ii) Hence, derive an expression for the period T of the oscillations, in terms of ρ, M, A and g.

$$\omega^2 = \left(\frac{\rho A g}{M}\right) \qquad \text{[M1]}$$

$$\Rightarrow T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{M}{A\rho g}} \qquad \text{[A1]}$$

[2]

(iii) The rod is given a small vertical displacement and then released. It oscillates, but damping soon causes it to come to rest after a few oscillations in the water.

State why the damping causes the oscillations to stop.

· (friction / drag acts on the rod and) loss of energy (causes amplitude decrease) [1]

.....

.....[1]

(iv) Information from Fig. 7.4 can be used to verify that the damping on the rod is exponential.

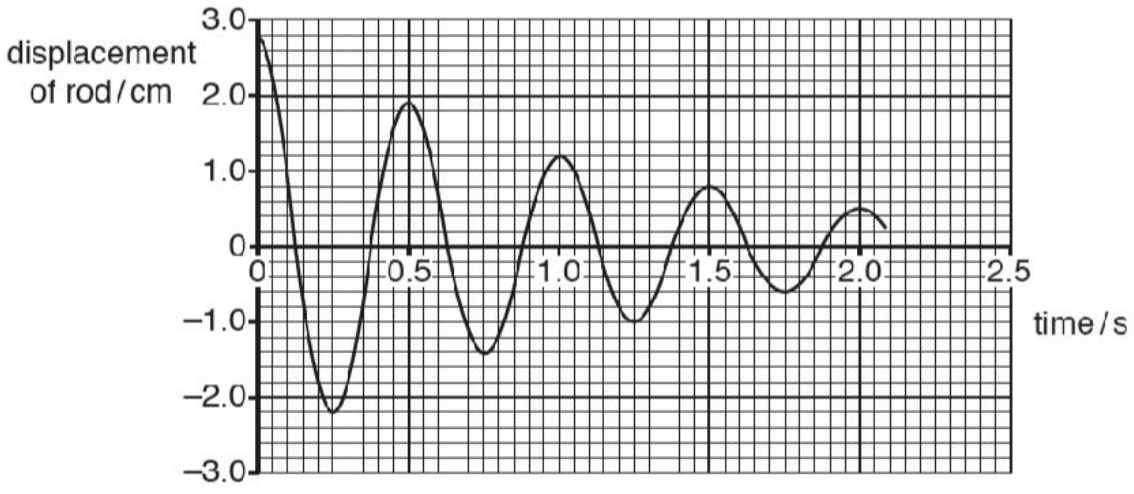


Fig. 7.4

			1.	Use Fig. 7.4 to complete the second column in the table of Fig. 7.5 to record the displacement at the times stated. [1]
			2.	Use the displacement data and the blank column in Fig. 7.5 to show that the rod exhibits light damping, where its amplitude decays exponentially with time. [1]
			3.	Use the grid in Fig. 7.6 to draw a graph that illustrates your answer. [2]

time / s	displacement / cm	
0.0		
0.5		
1.0		
1.5		
2.0		

2. calculate the values of [ln (displacement)] and show constant difference OR calculate the ratios of consecutive values and show constant value [1]

time / s	displacement / cm	ln displacement
0.0	2.8	1.03
0.5	1.9	0.642
1.0	1.2	0.182
1.5	0.8	−0.223
2.0	0.5	−0.692

1. 5 results correct [1]

only 1 error is 1 mark only

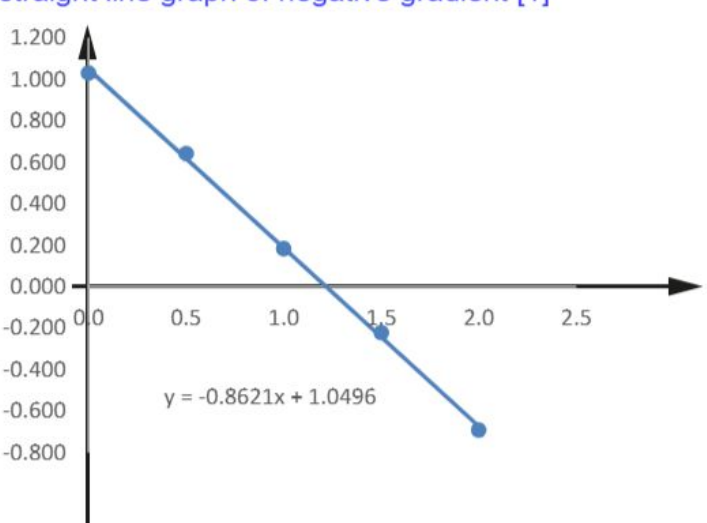
more than 1 error 0 marks

time / s	displacement / cm
0.0	2.8
0.5	1.9
1.0	1.2
1.5	0.8
2.0	0.5

Fig. 7.5

3. [ln (displacement)] against (time) fully labelled [1]

straight line graph of negative gradient [1]



OR: if plotting amplitude of rod against time [1],

show that the half-lives are the same from the graph [1]

Fig. 7.6

(d) Earthquake waves can produce forced oscillations in tall buildings. The buildings may collapse because resonance occurs.

Explain what is meant by the term *resonance* in this situation.

In this situation, the earthquake acts as the driving force and the building acts as the oscillating system. Resonance occurs when the frequency of the driver matches the natural frequency of the system [1], resulting in a maximum transfer of energy and a maximum amplitude of oscillation. [1]

(OR) Resonance occurs when the frequency of the external driver (the earthquake) matches the natural frequency of the oscillating system (the building) [1]. This leads to a maximum transfer of energy from the driver to the system, causing the building to oscillate with maximum amplitude. [1]

[2]

(e) Fig. 7.7 shows a tall building designed to prevent collapse during an earthquake. The building has a solid base which rests on a rubber bearing.

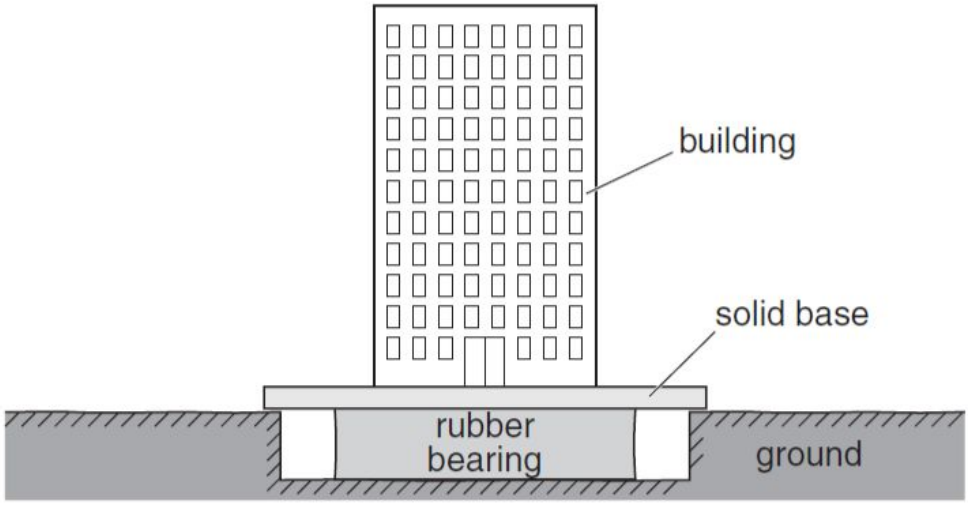


Fig. 7.7

Suggest why this design helps to prevent the building from collapse.

.....

The rubber bearing absorbs/dissipates kinetic energy from the earthquake (transforming it into thermal energy). [1]
OR: The rubber bearing increases the degree of damping of the system (the building).
Accept: Acts as a damper / absorber to absorb vibrations.

This reduces the maximum amplitude of the building's oscillations (at resonance). [1]
OR: It flattens the resonance curve, keeping the amplitude within safe limits.
Accept: Reduces the size/severity of the swings.

OR:

The rubber bearing increases the degree of damping within the oscillating system (the building) by absorbing and dissipating kinetic energy from the earthquake waves. [1]

In terms of a resonance graph, this heavier damping flattens the resonance peak, thereby drastically decreasing the maximum amplitude of the building's oscillations and preventing collapse. [1]

..... [2]

[Total: 20]